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31 January 1967
SUPERSEDING
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MILITARY SPECIFICATION

CARTRIDGE CASE, BASE PLUG, PROPELLANT CUP 40MM XM199

1. SCOPE

1.1 This specification is the applicable detail specification for Cartridge Case, Base Plug and Propellant Cup 40MM, XM199.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-10025	- Packaging, Packing and Marking for Interplant Shipping of Inert Ammunition Components, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1310

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2.1 (Cont'd)

DRAWINGS

Munitions Command

C10542348	- Case, Cartridge 40MM XM199
C10542350	- Base Plug
B10542352	- Propellant Cup
IEL-10542348	- Index of Inspection Equipment List for Case, Cartridge 40MM XM199

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 Metal defects.- All components and assemblies shall be free from cracks, splits, cold shuts, inclusions, porosity or any other defects.

3.3 Mechanical properties.- The mechanical properties specified on the applicable drawings and referenced specifications shall be uniform throughout each component.

3.4 Security of base plug to cartridge case.- The base plug or cartridge case threads shall show no evidence of failure when the applicable force is applied in an axial direction.

3.5 Protective coating.- Protective coating shall be in accordance with the requirements of the applicable drawing and specifications listed thereon.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

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3.6.2 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

15 Cartridge Cases (coated)
15 Cartridge Cases (uncoated)
15 Base Plugs (coated)
15 Base Plugs (uncoated)
15 Propellant cups

The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Lot formation.- A lot shall consist of cartridge cases, base plugs and propellant cups produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.2 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Case, cartridge, Drawing C10542348

	Defect	Method of Inspection
CRITICAL		
1.	Propellant flash hole(s) missing or obstructed	Visual
MAJOR		
101.	Diameter of sabot cavity not functionally concentric with outside body diameter	Gage
102.	Diameter of flange not functionally concentric with outside diameter	Gage
103.	Diameter of propellant cavity not functionally concentric with outside body diameter	Gage
104.	Rear face not perpendicular with outside diameter	Gage
105.	Pitch diameter of base plug thread not functionally concentric with outside diameter	Gage
106.	Pitch diameter of base plug thread, max.	Gage

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4.3.2.1 (Cont'd)

Defect		Method of Inspection
107.	Minor diameter of base plug thread, max.	Gage
108.	Diameter of flash holes	Gage
109.	Diameter of propellant cavity	Gage
110.	Diameter of body (outside)	Gage
111.	Diameter of sabot cavity	Gage
112.	Diameter of flange	Gage
113.	Depth of base plug cavity	Gage
114.	Depth from rear face to bottom of propellant cavity	Gage
115.	Length from rear face to bottom of sabot cavity	Gage
116.	Length of effective base plug thread, min.	Gage
117.	Thickness of flange	Gage
118.	Metal defective	Visual

MINOR

201.	Diameter of base plug thread undercut (if applicable)	Gage
202.	Depth of undercut forward of flange (if applicable)	Gage
203.	Width of undercut forward of flange (if applicable)	Gage
204.	Total length	Gage
205.	Radius of propellant cavity base	Gage
206.	Location of flash holes	Visual (see 6.1)
207.	Radii or chamfers missing or incorrect	Visual (see 6.1)
208.	Surface finish improper	Visual (see 6.1)
209.	Protective coating incorrect, inadequate or damaged	Visual (see 6.1)
210.	Marking missing, incorrect or illegible	Visual
211.	Burr	Visual
212.	Foreign matter	Visual

4.3.2.2 Cup propellant, Drawing B10542352

CRITICAL

None defined

MAJOR

101.	Diameter of body (outside) when assembled to post	Gage
102.	Thickness	Gage

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4.3.2.2 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Diameter of flange	Gage
202.	Length from front face to front of flange	Gage
203.	Distortion of front face max.	Visual (see 6.1)
204.	Radii or chamfers missing or incorrect	Visual (see 6.1)
205.	Burr	Visual
206.	Foreign matter	Visual

4.3.2.3 Plug, base, Drawing C10542350

CRITICAL		
1.	Primer flash hole missing or obstructed	Visual
MAJOR		
101.	Diameter of primer cavity not functionally concentric with pitch diameter of cartridge case thread	Gage
102.	Pitch diameter of cartridge case thread, min.	Gage
103.	Major diameter of cartridge case thread, min.	Gage
104.	Diameter of primer cavity	Gage
105.	Diameter of primer flash hole	Gage
106.	Depth of primer cavity	Gage
107.	Radius primer cavity base	Gage
108.	Threads damaged or not full	Visual
109.	Metal defective	Visual
MINOR		
201.	Diameter of wrench holes	Gage
202.	Depth of wrench holes	Gage
203.	Location of wrench holes	Gage
204.	Length total	Gage
205.	Radii or chamfers missing or incorrect	Visual (see 6.1)
206.	Surface finish improper	Visual (see 6.1)
207.	Protective coating incorrect, inadequate or damaged	Visual
208.	Burr	Visual
209.	Foreign matter	Visual

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4.3.3 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Finished aluminum components.- The sample size and acceptance criteria shall be as specified in the applicable material specification. Failure to comply with the requirements of the applicable drawing and specification shall be cause for rejection of the lot.

4.3.3.2 Base plug.

4.3.3.2.1 Security of base plug to cartridge case.- One cartridge case and one base plug from each eight (8) hours production or fraction thereof shall be subjected to this test. Failure of any cartridge case assembly to comply with 3.4 shall be cause for rejection of the lot. This is a destructive test. Parts so tested shall not be returned to the lot.

4.3.3.3 Protective coating.- A sample size of five (5) base plugs and five (5) cartridge cases per each eight (8) hours production shall be tested for compliance to the requirements of the applicable specification. Failure to comply with the requirements shall be cause for rejection of the lot represented by the sample.

4.3.4 Inspection equipment.- Index of Inspection Equipment List No. IEL 10542348 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format.)

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Finished aluminum components.- From the samples selected in accordance with 4.3.3.1.1 after final heat treatment, one tension test specimen conforming to the largest obtainable "F" type specimen as prescribed by Federal Test Method Standard No. 151, shall be taken in a circumferentially transverse direction forward of the web area. The criteria for performing the test shall be as prescribed by Federal Test Method Standard No. 151.

4.4.2 Base Plug.

4.4.2.1 Security of base plug to cartridge case.- The cartridge case supplied in accordance with 4.3.3.2.1 shall have the web bored out. The base plug shall be assembled to the cartridge case with a torque of 250 inch pound minimum to 400 inch pound maximum. The assembly will then be mounted on an approved device in such a manner the cartridge case will be held stationary. A 13,000 pound force will be applied to the internal face of the base plug. After the force has been released, the base plug and cartridge case will be examined for evidence of failure.

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4.4.3 Protective coating.- The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level A.- Not applicable

5.1.2 Level B.- Not applicable

5.1.3 Level C.- Packing and marking shall be in accordance with MIL-P-10025.

5.2 Data Card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1310-A048

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